

cocotb Clock helper

After triggers, you need a clock that runs while your test does other work

Period, start, and edge times

- The period is the full cycle length, low half, then high half
- With period ten and units nanoseconds
- Start runs alongside your async test; it does not block the rest of the coroutine
- Pair this with RisingEdge from the last module and you have a clean rhythm

Browser lab

Digital Design and Verification Platform

HomeTools

Home / Tools / cocotb Clock helper

INTERACTIVE TOOL

cocotb Clock helper

Sketch `Clock(dut.clk, period, units='ns').start()` — show period, generate posedge times at period, 2×period, 3×period. Starter: period 10 ns, edges at 10, 20, 30. Try preset 2 *us period* for units literacy.

Starter example: `Clock(dut.clk, 10, units="ns").start()` — posedges at 10, 20, 30 ns.

Load starter example

Challenges 0 / 22 cleared

Quiz: `period`: `Clock(dut.clk, period, ...)` period means...

☐ sim-time between successive posedges

☐ number of test failures allowed

☐ VCD file size in MB

☐ UVM verbosity level

Show hint

Check

Next

Idle

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

Core ideas

period

Time between successive posedges.

.start()

Background coroutine toggles clk.

edges

Posedges at period × n.

units

ns/ps/us label sim-time.

Browser lab starter

Real cocotb practice

- In the real cocotb track, sketch the helper on paper before you run a simulator
- Write Clock with dut dot clk, period ten, units ns, then start on its own line
- List the first three posedge times you expect
- Optional: open the lab's source sketch panel and read the comment block
- This module is clock-generation literacy, not a full UART testbench yet

Pitfalls to watch

- Do not confuse the Clock helper with manually flipping dut dot clk dot value every step
- A common miss is mixing up period with half-period
- Watch units too, ten nanoseconds and ten microseconds are not the same timeline
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load the starter and predict the third posedge before you reveal it
- On paper, write the Clock line and three edge times for period eight
- When you are ready